

Study of the aerodynamic interactions between engine and aircraft – AEGIS

(Bibliographic report)

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ABSTRACT:

This research focuses on the study of the aerodynamic impact of the installation of the propulsion system on a tube and wing type aircraft. The impact of the engine's position and size on lift and drag on a under wing configuration is studied before analyzing the over wing configuration. The results will then be implemented in the *Pacelab* preliminary design software.

I – Context

One of the objectives for future aircraft is reducing environmental impact. This translates into less fuel consumption and lower polluting gas emissions, but also reducing noise impact. As an example, one of the goals set in FlightPath 2050 by the European Commission, is to reduce CO₂ emissions by 75% per passenger per kilometer and perceived noise emission by 65%, relative to the capabilities of aircraft in the year 2000. In order to achieve these goals, the whole design of current aircraft is being rethought. The most promising leads for future aircrafts include distributed electric propulsion, rather than podded turbofans found on today's airplanes.

In 2016, the AEGIS research sponsorship was signed between ISAE-SUPAERO and SAFRAN, whose goal is to consolidate the innovative propulsion designs that will thrust future aircraft, while meeting the European Commission objectives.

In order to reduce fuel consumption, the current trend is to increase the bypass ratio of engines, which leads to an improvement in the propulsion efficiency. However, with a higher bypass ratio, larger is the diameter of the engine which can lead to poor aerodynamic performance.

The next generation of high bypass ratio engines, known as Ultra High Bypass Ratio engines (UHBR), calls for an investigation of the aerodynamical impact of their integration in the airplane, particularly the interferences that exist between the nacelle, pylon, wing and fuselage. New integration positions, different from the current under wing station, are being studied to minimize the additional drag produced by the installation of the engines.

Investigating the drag induced by the position and size of the engines is also important for the study of future distributed propulsion aircraft because on these aircraft, the engines will occupy a much larger surface and can thus induce greater drag.

II – Problem statement

The main difficulty that comes with evaluating the aerodynamic impact of the engines integration to the aircraft is the coupled nature of the interactions that exist between the different elements of the propulsion system. One must first define the additional drag components that are induced by the integration. The interference drag is the increment of drag that is produced when two elements are in close proximity: the total drag of the two coupled elements is greater than the sum of the drag of each individual component. The installation drag is the drag of the propulsion system (nacelle and pylon) plus the interference drag produced by these elements [7]. The objective is to minimize these additional drag components. The predominant interactions are nacelle-wing, pylon-wing and nacelle-fuselage.

Another difficulty is to determine the nature of this drag. Research shows that it is mainly viscous, wave and induced drag that is present [7].

In order to determine the additional drag produced by the engines installation and its

dependence on their size and position on the aircraft, wind tunnel experiments and numerical testing (CFD) have been carried out. Static pressure distribution around the nacelle, pylon and wing given by wind tunnel experiments allow to compare between a clean and motorized configuration and gives conclusive data regarding drag. CFD testing gives matching results to the wind tunnel experiments but gives particularly good predictions for lift impact [2].

III – First findings and future work

On a common underwing configuration, bibliographic research shows that the presence of the motorization creates an increase of 14% in drag [1] and a decrease in lift of 12% [3] compared to a clean wing configuration. More precisely, due to the presence of the nacelle, the shock wave present on the upper wing is moved forward, creating an increase in pressure on the upper wing [2] and an acceleration on the under wing near the pylon, resulting in lift loss. The presence of the nacelle also creates a supersonic region inboard of the nacelle, which increases wave drag [3].

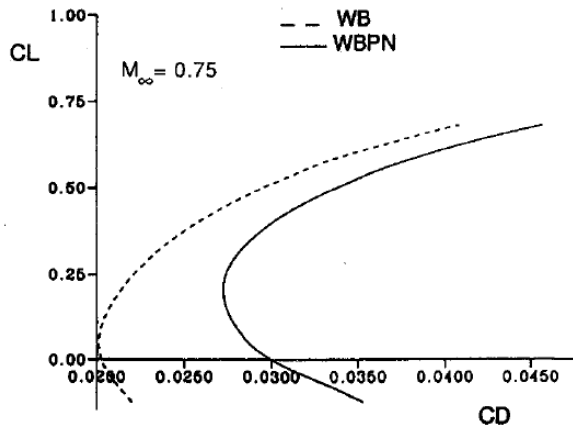


Fig.2 Lift-Drag Coefficient polar for a clean wing and motorized configuration [3]

Y position impact:

Research shows that the interference drag due to the nacelles decreases when the engines are installed at more outward wingspan positions [1] reducing the supersonic region inboard of the nacelle.

X position impact:

Research shows that a forward mounting is preferable for an underwing configuration [2][4] and is preferable further towards the trailing edge of the wing for an overwing configuration [5].

Z position impact:

Research shows that there is no significant correlation between the Z position of the nacelle and the aerodynamic impact [2] and will thus be ignored in the present study.

Nacelle size impact:

Lift loss is proportional to nacelle size [4] due to a more forward position of the shock wave on the upper wing.

Future work will focus on implementing the current findings of the Y, X and nacelle size impact in the *Pacelab* preliminary design software and will then focused on overwing engine mounting.

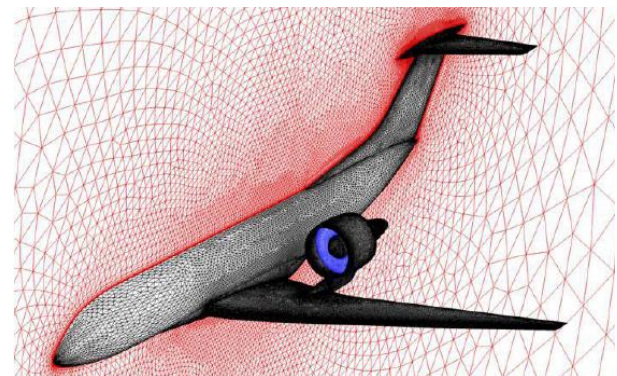


Fig.3 Unstructured mesh for a CFD study of an Over Wing UHBR configuration [5]

References

- [1] A.M Ingraldi, T.T Kariya, R.J Re. *Interference Effects of Very High-Bypass Ratio Nacelle Installations on a Low-Wing Transport*. ASME, 1991.
- [2] C.C. Rossow, H. Hoheisel. *Numerical Study of Interference Effects of Wing-Mounted Advanced Engine Concepts*. ICAS and AIAA, 1994.
- [3] C.C. Rossow, J.L. Godard, H. Hoheisel, V. Schmitt. *Investigation of Propulsion Integration Interference Effects on a Transport Aircraft Configuration*. Journal of Aircraft, Vol.31, No.5, Sept.-Oct. 1994.
- [4] R. Rudnik, C.C. Rossow. *Numerical Simulation of Engine/Airframe Integration for High-Bypass Engines*. ECCOMAS, 2000.
- [5] J.R. Hooker, A. Wick, C. Zeune, A. Agelastos. *Over Wing Nacelle Installations for Improved Energy Efficiency*. 31st AIAA Applied Aerodynamics Conference, June 24-27, 2013, San Diego, CA.
- [6] S.H. Berguin, D.N. Mavris. *Interactions in Over-Wing Nacelle Optimization*. AIAA 2013-0025.
- [7] B. Malouin, J.Y Trépanier, E. Laurendeau. *Installation and Interference Drag Decomposition via RANS far-field Methods*. Aerospace Science and Technology 54 (2016) p132-142.
- [8] M. Price, S. Raghunathan, R. Curran. *An Integrated Systems Engineering Approach to Aircraft Design*. Progress in Aerospace Sciences 42 (2006) p331-476.